

3. A present is placed at rest on a plane that is inclined, at a distance L from the bottom of the incline that makes an angle θ with the ground. At the bottom of the incline, the box is determined to have a speed v . If the box is instead released from a distance of $4L$ from the bottom of the incline, what will its speed at the bottom of the incline be?

$$v_2 = 2v_1$$

4. Imagine you are swinging a bucket full of water attached to a rope in a vertical circle

(a) Where along the trajectory is the tension on the rope the highest?

At the bottom because the tension directly opposes the weight of the bucket

(b) Why doesn't the water spill out of the bucket at the top of the circle? Give two explanations:

1. The normal force, centripetal force and gravitational forces cancel each other out. 2. The water is actually in free-fall, however so is the bucket. and since they are in movement together the water stays within the bucket

(c) What is the minimum speed required for the water to not spill out of the bucket?

$v = \sqrt{gR}$ where R is the radius of the circular path the bucket is rotating around.